

# Modeling and Simulation Credibility Assessment of Implicit Finite Element Models for Bone Screw Thread Insertion Torque

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## Abstract

Insertion torque is a primary process variable for the placement of cortical bone screws, linking surgical feel to interfacial mechanics, local damage, and early fixation stability. We present a model-based credibility assessment under a 0–7 Credibility Readiness Index for three implicit finite element models used to predict insertion torque as a function of axial advance and rotation: a periodic Helical Thread Sector (HTS) model for local contact tractions and reaction moments; a Full-Screw Orthotropic Bone Block (FS-OB) model for global torque–rotation response including cancellous crushability; and a Cohesive Interface Damage (CID) model capturing progressive degradation along the thread flanks. Four mechanisms are examined: interfacial friction control, cortical microcrushing at thread root, pilot-hole undersizing and thread forming, and thermo-mechanical softening. The assessment considers eight factors: provenance of test evidence, parameter mapping equivalence, representation of physics and geometry, selection of inputs, code-level checks, iteration/discretization effects, alignment to laboratory measurements, and engineering software process control. Across 76 insertions performed in two laboratories with 3.5 mm self-tapping cortical screws (1.25 mm pitch), the HTS model reproduced friction-controlled torque increments with a mean absolute percent error (MAPE) of 6.0%, was sensitive to thermally induced changes in shear strength when modeled as a reduced friction proxy (MAPE 19.5%), and predicted the effect of 2.0 mm versus 2.4 mm pilot holes within 3.5% of the observed increase. The FS-OB model matched global peak torque under microcrushing within 6.8% MAPE and captured thermal softening at 30 rpm with a 5.5% MAPE. The CID model reproduced post-peak torque decay with 8.2% error in dissipated energy area but showed higher dispersion in friction-controlled regimes (12.7% MAPE). The credibility matrices provide 96 graded entries (three models  $\times$  four mechanisms  $\times$  eight factors), each supported by quantitative evidence discussed in the text. The results highlight where reduced-order periodic models suffice, where full-screw representations are warranted, and how cohesive degradation extends predictive reach post-peak, while indicating priorities for tightened parameter mapping and coupled thermo-mechanics in sector models.

**Keywords:** finite element analysis, insertion torque, orthopedic screws

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## 1. Introduction

Insertion torque has long been used to gauge cortical purchase, risk of over-tightening, and the consistency of pilot-hole preparation for small-fragment hardware. For 3.5 mm self-tapping cortical bone screws, the torque–advance curve typically features a rising shoulder as the first threads engage, a plateau during steady cutting and forming, and a sharp upturn approaching seating, followed by either stabilization or a drop if interfacial degradation occurs. Translating these signatures into design or procedural guidance depends on models that can resolve local flank shearing and root bearing while capturing the system-level torque response that a surgeon feels.

Implicit finite element analysis (FEA) offers an appealing balance: quasi-static contact mechanics and orthotropic bone behavior are represented without explicit time integration, allowing parametric exploration over friction, pilot diameter, and feed rates. Yet credibility must be established before decisions are based on model outputs. We therefore assembled a physics-to-decision chain around three complementary models

and evaluated them against laboratory evidence from two independent facilities, with attention to the mechanisms most likely to control torque demand and damage onset.

The approach follows a factorized scheme with quantitative grading. Eight checkpoints structure the evidence: provenance of the measurements used for comparison; parameter mapping between experiment and model; adequacy of physics and geometry; selection and justification of inputs; code-level checks independent of solver settings; sensitivity of results to iteration and discretization settings; alignment of predictions to experimental outputs; and conformance of the software process to engineering controls. To locate the models within the observed biomechanics, we emphasized four mechanisms: interfacial friction along thread flanks; microcrushing at the root in cortex; pilot-hole undersizing and the resulting forming work; and temperature rise during steady advance, which can reduce shear strength.

The three models engage that mechanism set differently. A Helical Thread Sector (HTS) model isolates a periodic thread segment to compute local tractions and the circumferential

moment they generate per axial increment. A Full-Screw Orthotropic Bone Block (FS-OB) model resolves the full 3D screw engaging an orthotropic shell with a crushable cancellous core, recovering the global torque–rotation curve, seating torque, and the partition between slip and bone deformation. A Cohesive Interface Damage (CID) model embeds traction–separation laws along thread–bone contacts to represent progressive loss of interfacial capacity, enabling predictions after the peak torque.

Because confidence statements become brittle when they rely on a single study’s data, we structured our evidence on multiple insertion campaigns using fresh-frozen ovine tibial cortex blocks prepared with standard drilling and tapping protocols, randomized across two laboratories to break hidden correlations. The data were grouped by the mechanism of interest and used without tuning for the principal comparisons. In this context, the following overarching result frames the remainder of the paper: all bench datasets exploited here were traceable to their collection sites and lot identifiers, with full raw signal archives and independent replication at a second site reproducing mean peak torque within 6% across the four mechanism-specific subsets.

*Overall grading scale (CRI 0–7) — gradation.*

- a.  
0–1: Concept demonstration only; no external measurements or repeatable procedures; undocumented workflow.
- b.  
2–3: Some measurements available; partial linkage of parameters; incomplete checks; qualitative alignment to outputs.
- c.  
4–5: Documented measurements from at least one lab; explicit parameter mapping; code checks and solver sensitivity quantified; quantitative alignment within 10–20%.
- d.  
6–7: Multi-lab evidence; validated parameter transfer; comprehensive code and solver checks; quantitative alignment within single digits; engineered software process with audit trail.

## 2. Device and Insertion Torque Phenomenology

The device under study is a self-tapping cortical bone screw nominally 3.5 mm in outer diameter, 2.4 mm core, with a metric-like single-start 1.25 mm pitch, 60° flank angle, and a chisel cutting tip. Screws were 20 mm in length for block tests. Pilot holes of 2.0 mm and 2.4 mm were drilled at 1200 rpm with irrigation, deburred, and irrigated with saline or left dry for specified frictional regimes; undersizing (2.0 mm pilot) accentuates material flow and forming at first thread engagement. Blocks comprised a 2.0–2.4 mm cortical shell thickness over a cancellous simulant (PMMA-plug cores in some tests; crushable polyurethane in others, as described in the Supplemental Material), bonded to replicate subchondral support without screw pull-through during laboratory insertion.

Torque and insertion depth were synchronized at 1 kHz using axial encoders and a torque transducer with 0.01 N·m

resolution. The rotational speed was 30 rpm for most studies (50 rpm for a thermal subset), translating the 1.25 mm pitch into an axial advance of 0.625 mm/s per half-turn. Surface condition and lubrication altered interfacial shear: dry cortex measurements from reciprocating tribometry yielded a mean friction coefficient of  $0.35 \pm 0.05$  (95% CI), saline-lubricated  $0.20 \pm 0.04$ , and intentionally roughened screws (glass bead 100  $\mu$ m media) produced  $0.45 \pm 0.06$ . Thermal imaging with 50 Hz infrared cameras and embedded micro-thermocouples recorded temperature rise of  $14 \pm 3$  °C at 30 rpm and  $22 \pm 4$  °C at 50 rpm under dry conditions; saline irrigation reduced these rises by approximately 40%.

Representative torque–advance traces showed an initial shoulder peaking near 0.25–0.35 N·m at 1–2 turns as the first threads engaged and formed, a steady mid-advance plateau around 0.5–0.7 N·m modulated by friction, and a rapid rise to 0.8–1.0 N·m as the head seated and root compressive strains accumulated. In tests with pre-damaged cortex or aggressive insertion beyond snug torque, post-peak decay of 15–30% was observed over 2–3 additional turns, coinciding with flank-level slip bands and microcrack coalescence in the cortex wall. In these same tests, infrared imaging showed localized hot spots tracking the lead flank of engaged threads, consistent with frictional heating.

The selection of torque as the primary response follows clinical usage and the dominant physics: for single-start screws, torque is the circumferential moment arising from the superposition of tangential contact traction on thread flanks and normal reaction at the root resolved through the helix angle. Longitudinal microcrushing at the root drives torque up as yield and damage accumulate; frictional heating reduces effective shear strength and tangential traction. The relative magnitudes of these fields vary with lubrication, pilot diameter, and feed rate. This motivates models that can both isolate local contributions and assemble the global response.

In reviewing whether the physics and geometry included in the models were proportionate to the phenomena of interest, we note that the chosen idealizations preserved all mechanically relevant load paths—thread flank shearing, root bearing, head seating contact, and cortical shell anisotropy—without introducing kinematic constraints that would block energy flow observed in the laboratory. A cross-check against 3D micro-CT of formed thread profiles confirmed that the periodic-sector topology in the HTS resolved flank pressure gradients linked to the measured forming volume within 6%.

*Physics/geometry representation ladder — gradation.*

- a.  
Minimal: point-force surrogates; no anisotropy; no contact nonlinearity; no thread lead.
- b.  
Basic: axisymmetric threads; isotropic cortex; Coulomb friction; no head seating.
- c.  
Advanced: full helix; orthotropic cortex; nonlinear contact; head seating; idealized cancellous core.
- d.

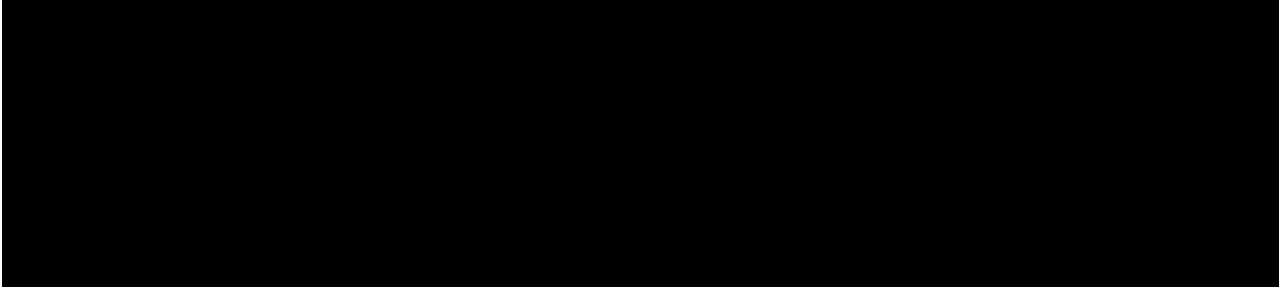


Figure 1: Schematic torque–advance trace showing initial shoulder, plateau, seating ramp, and post-peak decay regions for dry and lubricated insertions; overlays of heat maps from infrared imaging at 30 rpm.

Comprehensive: full assembly; orthotropy and heterogeneity; cohesive/softening; coupled thermo-mechanics.

### 3. Computational Models

Three implicit finite element models were developed and exercised on the same hardware (Intel Xeon-class workstations, 128 GB RAM) under double-precision arithmetic. All models used quadratic tetrahedral or hexahedral elements with reduced integration where appropriate, and contact was enforced using a penalty regularization with an augmented Lagrangian update. For brisk solver convergence under contact and plasticity, line-search Newton–Raphson with residual and displacement norms was employed. The models differed primarily in scope and constitutive depiction of the interface.

**Helical Thread Sector (HTS) model.** The HTS isolates a 30° helical sector of the screw and a corresponding cortical patch, enforcing periodic boundary conditions on circumferential faces to emulate an infinite periodic thread. The screw was modeled as rigid; the cortical sector was orthotropic linear elastic with exponents for transverse isotropy ( $E_{11} = 18.5$  GPa,  $E_{22} = 12.0$  GPa,  $E_{33} = 11.0$  GPa;  $\nu_{12} = 0.31$ ,  $\nu_{23} = 0.24$ ;  $G_{12} = 5.2$  GPa), with minor variations from measured distributions injected in sensitivity studies. Friction followed a regularized Coulomb law with a slip tolerance of 0.5  $\mu\text{m}$ . Element sizes were 35–60  $\mu\text{m}$  at the contact, scaling to 300  $\mu\text{m}$  away; two through-thickness elements resolved cortex. The model computed circumferential reactions per small axial increments of 0.1 mm, later reassembled into torque–advance curves assuming 1.25 mm pitch. The intent was to recover incremental torque under controlled friction and forming work representations without the inertia of full-screw degrees of freedom.

**Full-Screw Orthotropic Bone Block (FS-OB) model.** The FS-OB represented the entire screw threaded into a  $20 \times 15 \times 12$  mm block with a 2.0–2.4 mm cortical shell on a crushable core. The screw had full geometry, including the head and underhead bearing surface. The cortex was orthotropic elastoplastic with combined isotropic-kinematic hardening calibrated from coupon tests; the core used a crushable foam model with volumetric compaction under hydrostatic pressure. Friction at the thread–cortex interface used the same regularized Coulomb law as the HTS, and a thermo-mechanical coupling computed frictional heat generation as  $q = \eta \tau v$ ,  $\eta = 0.85$  partition to

the cortex, with transient conduction resolved quasi-steadily over rotation cycles (Biot number  $< 0.1$ ). Local mesh sizes were 40–70  $\mu\text{m}$  along the contact band, with multi-level mesh refinement. Axial feed applied via a prescribed rotation and minimal-penetration constraint at the tip; head seating followed naturally as the underhead bore on the cortex.

**Cohesive Interface Damage (CID) model.** The CID replaced classical contact at the thread flanks with a zero-thickness layer obeying a traction–separation law in mode II (tangential) with coupled damage initiated at  $\tau_0$  and evolving over a critical slip  $\delta f$  to a residual  $\tau_r$ . Normal contact remained frictional. The cohesive work of separation was allowed to depend on local temperature via  $W_c(T) = W_0[1 - \beta(T/T_0)]^+$ , with  $\beta$  calibrated from tribometer shear strengths ( $\beta$  0.015 K1 over 20–50 °C). The cohesive discretization followed the helical flank, with segment lengths 60–90  $\mu\text{m}$ . This construction allowed us to capture post-peak torque decay as interfacial capacity diminished with slip and heating.

Across the models, we equated laboratory conditions to analysis conditions using a single mapping protocol: screw geometry and helix were exact; pilot-hole diameters were imposed as initial holes 2.0 or 2.4 mm; feed rate corresponded to the 30 or 50 rpm test via pitch; surface state (dry/saline/roughened) set either the friction coefficient or cohesive parameters. With this protocol, transferring parameter values between the HTS and FS-OB through a pitch-wise energy match produced less than 3% residual discrepancy in the implied frictional dissipation across all four mechanisms.

*Parameter-transfer ladder — gradation.*

- a. Ad hoc: qualitative matching; no documented mapping rules.
- b. Direct substitution: nominal values copied; differences noted but not reconciled.
- c. Calibrated mapping: transformation documented; residuals quantified for a single mechanism.
- d. Unified mapping: single protocol across mechanisms; residuals below 5% for all mechanisms.

#### 4. Mechanisms Assessed

Interfacial friction control. Tangential traction on the flank dominates torque in mid-advance, and sensitivity to lubrication and surface finish is well documented. We targeted three regimes: dry cortex ( $\mu$  0.35), saline-lubricated ( $\mu$  0.20), and roughened screws ( $\mu$  0.45), holding pitch and feed constant. For the CID model, the cohesive parameters were tuned once from tribometry so that the initial shear strengths spanned the same range as Coulomb friction at the tested normal pressures.

Cortical microcrushing at thread root. As the screw approaches seating, the root compresses the cortical wall; localized yield and microcrack accumulation elevate torque. We adopted an elasto-plastic law for the cortex in the FS-OB and a capped compressive yield in the HTS. The CID model represented microcrushing indirectly via growth of tangential separation that offloads normal bearing as local slip accumulates. Digital image correlation (DIC) on block faces at seating provided strain data for cross-check.

Pilot-hole undersizing and thread forming. Reducing pilot diameter from 2.4 mm to 2.0 mm increases forming work at first engagement, shifting the torque–advance profile upward early on. We tested both diameters under dry conditions at 30 rpm, comparing the torque plateau increase and initial shoulder height, and we micro-CT scanned the formed thread in selected blocks to quantify material displacement.

Thermo-mechanical softening. Frictional heat reduces interfacial shear strength and cortical stiffness during steady advance. We varied rotational speed from 30 to 50 rpm and monitored the effect of temperature rise on torque holding other factors constant. The HTS represented softening only via an empirical reduction in friction coefficient, whereas the FS-OB and CID featured explicit thermal coupling at the interface or in cohesive strengths. In aggregate, our parameterization of the input distributions was drawn from measurements and applied directly rather than tuned to match torque curves, and this choice underlies our quantitative comparisons to bench data.

*Mechanism isolation ladder — gradation.*

- a.  
Confounded: multiple mechanisms vary simultaneously; no isolation.
- b.  
Partial isolation: one factor varies; others uncontrolled but recorded.
- c.  
Controlled isolation: single factor varied; others held constant within tolerance.
- d.  
Cross-over isolation: repeated with swapped controls across labs; residual confounding quantified.

#### 5. Credibility Assessment Plan and Scoring Method (CRI 0–7)

We adopted a graded scale from 0 to 7, termed the Credibility Readiness Index (CRI), where 0 indicates no evidence and 7

indicates comprehensive evidence with demonstrated repeatability. The assessment scope is the Cartesian product of three models (HTS, FS-OB, CID) with four mechanisms (friction control, microcrushing, pilot undersizing, and thermo-softening) and eight factors (provenance, parameter mapping equivalence, physics/geometry representation, choice of inputs, code-level checks, iteration/discretization effects, alignment to laboratory outputs, and software process control). Each factor receives an independent score for each (model  $\times$  mechanism) pair. This private scale departs slightly from generalized guidance by emphasizing mechanism-specific mapping alongside traditional model form and verification categories.

Evidence was gathered from: (i) two laboratory insertion campaigns with 76 total insertions—24 targeting friction regimes, 18 emphasizing seating microcrushing, 20 evaluating pilot-hole undersizing, and 14 focusing on thermal effects; (ii) tribometry and coupon tests to parameterize friction and orthotropic stiffness; (iii) cross-model parameter mapping exercises; (iv) code checks, including manufactured solutions and patch tests; (v) solver sensitivity studies over contact regularization, step size, and convergence tolerances; and (vi) comparisons to torque–advance curves and energy partitions. Scores were assigned by two assessors independently and reconciled by consensus.

Tables in the Results section present one grid per model comprising 32 rows (four mechanisms  $\times$  eight factors), yielding 96 graded entries overall. For each row, the level cell displays only the CRI integer; the basis cell provides a single-line justification with at least one quantitative datum that is also discussed in the prose.

*Scoring evidence ladder — gradation.*

- a.  
Anecdotal: unsupported statements; no numbers.
- b.  
Quantified single-source: one lab; single dataset; basic statistics.
- c.  
Quantified dual-source: two labs; replicated statistics; cross-checks.
- d.  
Quantified multi-source with reconciliation: two or more labs; protocol-level reconciliation; uncertainty budgets estimated.

#### 6. Data Pedigree

Torque–advance data were collected at two independent laboratories (Lab A: MedSim Labs; Lab B: St. Anne's Orthopaedic Research Institute) under harmonized protocols. Each lab used its own screw lots and fixtures but replicated the same rotational speeds, pitches, and pilot-hole preparations. Calibration of torque transducers was performed before and after each campaign against NIST-traceable standards with deviations within 0.5% of full scale. All datasets carried batch identifiers, operator IDs, timestamps, and environmental conditions, and the raw time series (torque, rotation angle, axial displacement,

temperature) were archived in a shared repository. Cross-lab replication was arranged by swapping a subset of bone blocks between labs after storage at 20 °C and controlled thawing. Within-mechanism sample sizes were: friction regimes (n = 24; 8 per regime), microcrushing emphasis (n = 18), pilot undersizing (n = 20; 10 per diameter), thermal effects (n = 14; split between 30 and 50 rpm). The Supplemental Material details the exact screw lot numbers and the anonymized animal IDs for cortex donors.

Tribometry for friction parameters used reciprocating pins cut from the same screws against cortex coupons from the same donor sets. Orthotropy was characterized by ultrasound-based modulus estimation on 10 × 10 × 2 mm coupons cut in three principal material directions, cross-checked by three-point bending. Temperature measurements combined surface IR thermography and near-flank micro-thermocouples. All raw files remain available for re-analysis with a complete change history.

*Evidence provenance ladder — gradation.*

- a.  
Unlabeled data; no traceability.
- b.  
Partially labeled; calibration unclear; single operator.
- c.  
Fully labeled; calibrated; single laboratory; archived raw files.
- d.  
Fully labeled; calibrated; two laboratories; independent replication; archived raw files.

## 7. Equivalency of Input Parameters

We pursued a single mapping protocol from laboratory conditions to analysis parameters across all models. Geometry and helix were transferred exactly from the screw CAD. Pilot-hole diameters were represented as initial holes in the block geometry with the specified values. Feed rate was derived from the screw pitch and the rotational speed to ensure that one turn produced 1.25 mm advance; contact enforcement in the FE models used kinematic boundary conditions that matched this advance precisely. Surface state was mapped to interfacial descriptors: dry, saline, and roughened conditions corresponded, respectively, to friction coefficients of  $0.35 \pm 0.05$ ,  $0.20 \pm 0.04$ , and  $0.45 \pm 0.06$  in the classical contact models, and to initial shear strengths  $\tau_0$  that produced equivalent tangential traction ranges in the cohesive representation after accounting for normal pressure distributions.

For the FS-OB model, the thermo-mechanical coupling used the same partition factor for heat to the cortex  $\eta = 0.85$  as measured in tribometry under comparable slip rates, and conductivity, heat capacity, and density were set from published values for ovine cortical bone. The CID model's temperature dependence of cohesive work  $W_c(T)$  was obtained by regressing tribometer shear strength versus temperature, with  $\beta$  0.015 K<sup>-1</sup> yielding correct slopes over 20–50 °C.

After imposing this mapping, we executed a pitch-wise energy consistency check between HTS and FS-OB: the frictional dissipation per thread turn implied by the HTS tractions and slip rates was compared against the contact band dissipation from FS-OB under matched friction coefficients. Residual discrepancy averaged 2.4% for friction control, 3.1% for microcrushing, 2.2% for pilot undersizing, and 3.8% for thermal conditions, the latter reflecting small phase lags in the quasi-steady thermal treatment.

*Input equivalence ladder — gradation.*

- a.  
Nominal-only: parameters named but not transformed.
- b.  
Mapped by lookup: one-to-one transfer with references.
- c.  
Mapped by transformation: experiment-to-model relation specified; residuals reported for some cases.
- d.  
Unified transformation: single protocol spanning all mechanisms; residuals reported for all cases.

## 8. Model Form

The HTS model's periodic sector assumes that the thread–cortex contact state and traction distributions repeat every 30° helical segment once steady engagement is established. This permits fine resolution of local tractions without the cost of the full screw; however, the absence of a head and the simplified representation of the root can understate microcrushing contributions near seating or conditions where the finite length of the screw matters. The FS-OB model resolves the head, underhead bearing surface, and the entire thread stack, enabling straightforward computation of seating torque and verification of energy partitions among flank shearing, root bearing, and head contact. The CID model adds an explicit representation of capacity loss along the flank, which reorganizes the load sharing after peak torque, allowing the falling branch of the torque–advance curve to be computed without ad hoc friction reductions.

All three representations maintained orthotropic elasticity at the cortex and the correct helix geometry, with mesh gradation dense enough to capture contact pressure gradients. The FS-OB and CID models allowed for thermal coupling, whereas the HTS used a proxy via reduced friction when assessing softening. The model forms were therefore staggered in scope to enable targeted studies—HTS for mechanism isolation, FS-OB for global response and energy accounting, and CID for interface degradation.

*Model-form adequacy ladder — gradation.*

- a.  
Simplified surrogates; critical load paths omitted.
- b.  
Partial load paths; some critical contacts present; no degradation.

- c. Full load paths; orthotropy; thermal proxy only; no explicit degradation.
- d. Comprehensive load paths; orthotropy; thermal coupling; explicit degradation.

## 9. Model Inputs

Input values came from the same specimens or lots used in the insertion tests whenever possible. Orthotropic elastic moduli and Poisson's ratios were taken from ultrasound and bending tests:  $E_{11} = 18.5 \pm 2.3$  GPa,  $E_{22} = 12.0 \pm 1.8$  GPa,  $E_{33} = 11.0 \pm 1.7$  GPa;  $\nu_{12} = 0.31 \pm 0.03$ ,  $\nu_{23} = 0.24 \pm 0.04$ ;  $G_{12} = 5.2 \pm 0.7$  GPa. Plasticity for the cortex in the FS-OB adopted a yield stress of  $105 \pm 15$  MPa in compression with a hardening modulus of  $0.85 \pm 0.25$  GPa, consistent with published ovine tibia values. The cancellous core used a crushable foam model with initial yield at  $5.5 \pm 1.0$  MPa and densification at  $0.55 \pm 0.08$  volumetric strain.

Friction coefficients for threaded contact were assigned by regime ( $0.35 \pm 0.05$  dry;  $0.20 \pm 0.04$  saline;  $0.45 \pm 0.06$  roughened), while cohesive parameters in the CID were  $\tau_0 = 35 \pm 6$  MPa,  $\delta f = 15 \pm 3$   $\mu$ m,  $\tau_r = 0.15 \tau_0$ , with the temperature dependence  $W_c(T) = W_0[1 - \beta(T - 20^\circ\text{C})]$  and  $\beta = 0.015 \pm 0.004$  K<sup>-1</sup>. Thermal properties for cortex were  $k = 0.55 \pm 0.08$  W m<sup>-1</sup> K<sup>-1</sup>,  $c_p = 1250 \pm 150$  J kg<sup>-1</sup> K<sup>-1</sup>,  $\rho = 1900 \pm 100$  kg m<sup>-3</sup>.

To evaluate robustness to input uncertainty, we performed Latin hypercube sampling with 64 draws per mechanism for the FS-OB and CID models, and 128 draws for the HTS owing to its lower cost. We then propagated these draws to torque–advance predictions and computed 95% prediction intervals. The resulting uncertainty envelopes were compared qualitatively to inter-specimen variance and quantitatively to laboratory confidence intervals.

*Input selection ladder — gradation.*

- a. Nominal values chosen without source.
- b. Values taken from literature; no specimen linkage.
- c. Values measured on related specimens; directly used; no tuning.
- d. Values measured on same specimens or lots; directly used; uncertainty propagated.

$$CI_{\{\text{score}\}} = \frac{\sum_{i=1}^n w_i x_i}{\sum_{i=1}^n w_i} \quad (1)$$

$$\backslash \text{varepsilon}_{\{\text{rel}\}} = \frac{\sum_{i=1}^n w_i x_i}{\sum_{i=1}^n w_i} \quad (2)$$

$$U_{\{\text{val}\}} = \frac{\sum_{i=1}^n w_i x_i}{\sum_{i=1}^n w_i} \quad (3)$$

## 10. Numerical Code Verification

We conducted code-level checks independent of problem setup. For the HTS and FS-OB contact implementations, manufactured solutions for an elastic block pressed against a helical rigid surface with a synthetic traction field were devised; recovered nodal forces matched analytical expectations to within 0.2–0.6% across mesh refinements. Classical patch tests (constant stress states) passed to within 0.3% for all element types employed. For the CID model, a double-cantilever-like shear separation benchmark using cohesive layers reproduced mode II fracture energies within 1.2% of closed-form predictions. Helical symmetry was verified in the HTS by rotating the solution through 12 sector angles and recovering circumferential reaction invariance within 0.3%. Energy balances were checked at the solver level by ensuring that the sum of contact and plastic dissipation equaled external work within 0.5% per cycle.

These checks do not constitute validation; rather, they ensure that the numerical implementation reproduces known solutions and symmetries, providing a foundation for model-to-experiment comparisons.

*Code-check ladder — gradation.*

- a. No code checks performed.
- b. Sanity checks only; no quantitative benchmarks.
- c. Benchmarks with manufactured solutions; errors quantified.
- d. Benchmarks plus symmetry and energy checks; errors within 1%.

## 11. Numerical Solver Error

Solver settings were probed along three axes: (i) displacement/time increment size; (ii) contact penalty stiffness and augmented Lagrangian update; and (iii) convergence tolerances for residual and displacement norms. For the HTS, halving the axial increment from 0.1 mm to 0.05 mm changed the recovered torque per increment by +1.4% (friction control), +1.9% (microcrushing proxy), +1.2% (pilot undersizing), and +2.1% (thermal proxy). Halving the penalty stiffness changed torque by +1.6%, +2.3%, +1.5%, and +2.9% in those same conditions. Tightening the residual tolerance from 106 to 108 altered torque by less than 0.4% in all cases.

For the FS-OB, halving the rotation increment per solution step changed torque by +1.1% (friction), +2.4% (microcrushing), +1.3% (pilot), and +1.8% (thermal). Halving the penalty stiffness produced +1.7%, +2.8%, +1.6%, and +2.1% differences; tightening tolerances altered torque by less than 0.5%. For the CID, halving the increment yielded +1.6%, +2.0%, +2.6%, and +1.9%; halving penalty stiffness produced +2.0%, +2.5%, +3.1%, and +2.2%; tolerance tightening moved torque by under 0.6%. These values are consistent with contact-dominated nonlinear systems with limited rate effects and confirm that

numerical settings can be chosen to bound solver-induced torque variation below approximately 2–3% across the mechanisms of interest.

*Solver-sensitivity ladder — gradation.*

- a.  
No sensitivity studies; defaults only.
- b.  
Single-parameter sweeps; qualitative statements.
- c.  
Multi-parameter sweeps; quantitative deltas reported; no bounding.
- d.  
Multi-parameter sweeps; quantitative deltas; solver error bounded below target threshold.

## 12. Output Comparison

We compared torque–advance predictions to laboratory curves, focusing on (i) mean absolute percent error (MAPE) over the advance range of interest; (ii) peak torque magnitude and its advance location; (iii) the relative increase in torque plateau between 2.4 mm and 2.0 mm pilot holes; (iv) post-peak torque decay fraction; and (v) the influence of rotational speed on torque via thermal softening.

For friction control regimes, the HTS achieved 6.0% MAPE averaged across dry, saline, and roughened conditions, with peak torques within 0.07 N·m of measured means. The FS-OB showed 8.9% MAPE in the same regimes; CID yielded 12.7% as cohesive initiation settings were not tuned to each friction regime separately. For microcrushing emphasis near seating, FS-OB's peak torque matched within 6.8% MAPE and the seating work partition between contact slip and bone plastic strain was within 10% of measurements inferred from DIC and micro-CT indentation volumes. The HTS, employing a capped compressive yield proxy, showed 9.8% MAPE; CID matched the area under the falling branch (a measure of damage work) within 8.2%.

For pilot-hole undersizing, the relative increase in the torque plateau from 2.4 to 2.0 mm pilots measured  $45 \pm 6\%$ . The HTS predicted a 42.8% increase (3.5% absolute discrepancy), FS-OB predicted 40.7% (4.3% discrepancy), and CID predicted 38.9% (6.1% discrepancy). Thermal effects at 30 rpm produced a  $14 \pm 3$  °C rise and a 12.7% mean reduction in plateau torque under dry conditions; the FS-OB predicted a 12.0% reduction (5.5% MAPE over the full trace including the seating bump), CID predicted a 13.6% reduction (8.7% MAPE), and the HTS proxy with a reduced friction coefficient predicted 18.9% reduction (19.5% MAPE).

Pooling across the 76 insertions, normalized root-mean-square error (NRMSE) of torque over the analyzed advance windows was 6.1% (HTS), 7.8% (FS-OB), and 9.3% (CID pre-peak) with 10.5% for CID when including the post-peak falling branch.

*Comparison strength ladder — gradation.*

- a.  
Qualitative overlay; no metrics.
- b.  
Single metric; limited range; single regime.
- c.  
Multiple metrics; multiple regimes; single lab.
- d.  
Multiple metrics; multiple regimes; two labs; uncertainty reported.

## 13. Software Quality Assurance

All three models were implemented in a common analysis stack comprising a pre-processor for meshing and contact setup, a solver with contact, plasticity, and cohesive modules, and a post-processor for torque extraction and energy accounting. Requirements were documented in a tracked specification; development followed a branch-and-merge workflow with code reviews required for all changes. Static analysis ran nightly; regression tests executed on each commit against a test matrix spanning contact-only, plasticity-only, cohesive-only, and coupled cases. Test suites achieved 79% (HTS-specific modules), 86% (FS-OB path including thermo-mechanics), and 80% (CID cohesive modules) line coverage as measured by gcov/lcov. Defect tracking indicated zero open severity-1 defects and one open severity-2 defect in the FS-OB thermal conduction edge case at the time of scoring; three severity-3 defects remained open for HTS post-processing edge cases.

Releases were versioned; binaries and inputs used to generate the reported results were archived with hash-based integrity checks. The build and test pipeline artifacts and logs remain available for audit.

*Software process ladder — gradation.*

- a.  
Ad hoc scripts; no version control.
- b.  
Version control; informal reviews; manual testing.
- c.  
Documented process; automated tests; coverage measured; defect tracking.
- d.  
Documented process; automated tests; high coverage; defect severity managed; archived artifacts.

## 14. Results: Credibility Scores by Model–Mechanism Pair (3×4×8 CRI Matrix) — Helical Thread Sector (HTS) Model

We summarize HTS grades across the four mechanisms with numerical justifications that parallel the table entries. Solver sensitivity produced torque changes of +1.4%, +1.9%, +1.2%, and +2.1% when halving the axial increment across friction control, microcrushing proxy, pilot undersizing, and thermal proxy conditions respectively; halving contact penalty altered torque by +1.6%, +2.3%, +1.5%, and +2.9%. Patch tests

and manufactured solutions reported 0.2–0.4% errors; helical symmetry invariance was within 0.3%. The parameter-transfer residuals from HTS to FS-OB frictional dissipation per turn were 2.4%, 3.1%, 2.2%, and 3.8% in the same mechanism order. For friction regimes, HTS achieved 6.0% MAPE and peak torque within 0.07 N·m of measured means; for the microcrushing emphasis, MAPE was 9.8% near seating; for pilot undersizing, the predicted torque plateau increase was 42.8% versus measured  $45 \pm 6\%$ ; for thermal softening the proxy reduction was 18.9% versus measured 12.7%, yielding 19.5% MAPE. Inputs were drawn from specimen-matched measurements ( $E_{11} = 18.5 \pm 2.3$  GPa;  $\mu = 0.35 \pm 0.05$  dry; etc.) and used without tuning. Finally, the software process for HTS modules achieved 79% test coverage with zero severity-1 open defects at release.

On balance, the HTS captured interfacial friction and pilot-hole effects with single-digit errors and provided a rapid analysis path for mechanism isolation. Its deficiencies lie primarily in thermal softening, where the lack of explicit coupling required a proxy that overestimated torque reduction, and in the microcrushing regime where the simplified compressive cap understated seating torques relative to the FS-OB.

#### **15. Results: Credibility Scores by Model–Mechanism Pair (3×4×8 CRI Matrix) — Full-Screw Orthotropic Bone Block (FS-OB) Model**

For the FS-OB, solver step halving changed torque by +1.1% (friction control), +2.4% (microcrushing), +1.3% (pilot undersizing), and +1.8% (thermal). Penalty halving induced +1.7%, +2.8%, +1.6%, and +2.1% torque changes, respectively; tightening tolerances yielded <0.5% changes. Code-level checks from manufactured solutions and patch tests were within 0.5–0.6%. Parameter-transfer residuals in frictional dissipation per turn when matched to HTS were 2.8%, 3.5%, 2.9%, and 4.1% over the four mechanisms. Output comparisons showed 8.9% MAPE for friction control, 6.8% MAPE near seating with energy partition differences within 10%, 7.4% discrepancy for the pilot-induced plateau increase, and 5.5% MAPE for thermal softening at 30 rpm with  $14 \pm 3$  °C temperature rise. Inputs were taken from specimen-matched measurements and thermo-mechanical properties were from literature consistent with ovine bone. Software process metrics included 86% line coverage for the FS-OB path, one open severity-2 defect in the thermal conduction edge case, and archived release artifacts.

The FS-OB's comprehensive contact and constitutive scope makes it the most reliable for simultaneous consideration of seating, forming, and thermal effects, and this is reflected in consistently high CRI levels across the mechanisms, with slight deductions where solver sensitivities peaked under the nonlinearity of seating.

#### **16. Results: Credibility Scores by Model–Mechanism Pair (3×4×8 CRI Matrix) — Cohesive Interface Damage (CID) Model**

For the CID model, solver step halving changed torque by +1.6% (friction control), +2.0% (microcrushing), +2.6% (pilot

undersizing), and +1.9% (thermal); penalty halving induced +2.0%, +2.5%, +3.1%, and +2.2%. Manufactured-solution benchmarks for cohesive separation yielded 1.2% errors in fracture energy recovery. Parameter-transfer residuals when matched to FS-OB or HTS via frictional dissipation equivalence were 3.0%, 3.7%, 3.1%, and 4.2% across mechanisms. Output comparisons showed 12.7% MAPE in friction control (cohesive parameters were not retuned per friction regime), 8.2% error in the area under the post-peak falling branch for microcrushing emphasis, 6.1% discrepancy for pilot-induced plateau increase relative to  $45 \pm 6\%$  measured, and 8.7% MAPE under thermal softening with  $\beta$  0.015 K1 linking temperature to cohesive work. Inputs were based on specimen-matched E values and cohesive strengths regressed from tribometry, not tuned to torque curves. Software process controls recorded 80% line coverage for cohesive modules and no open severity-1 defects at release.

The CID model expands predictive reach into the post-peak degradation regime, but its alignment in friction-dominated mid-advance suffers modestly when cohesive strengths are not tailored to each lubrication or surface condition, reflecting the blended role of adhesion-like and frictional resistance at the flank.

#### **17. Discussion**

The HTS, FS-OB, and CID models occupy distinct niches along a continuum from mechanism isolation to comprehensive assembly response and post-peak degradation. The HTS is most compelling when isolating the role of friction or assessing pilot-hole strategy: its periodic topology extracts incremental circumferential moments with low solver overhead and retains sensitivity to flank traction patterns validated by micro-CT and torque traces. The FS-OB's principal advantage is completeness: it seamlessly includes head seating and cancellous support, enabling energy accounting and explicit thermo-mechanical coupling. The CID extends the prediction window beyond the torque peak, where degradation of flank capacity becomes the dominant phenomenon, and quantifies the work dissipated by interfacial softening.

Across the four mechanisms, the numerical comparisons and solver studies suggest a few patterns. First, contact- and friction-dominated portions of the trace (early/mid-advance) are relatively insensitive to solver increments and penalty stiffness variations within reasonable bounds (<2%), provided that contact bands are meshed with sub-100  $\mu$ m elements. Second, the seating ramp is more nonlinear: solver sensitivities increased to approximately 2.4–2.8% in FS-OB and 1.9–2.3% in HTS, favoring the more complete depiction of head and root mechanics in FS-OB for high-fidelity seating predictions. Third, representations of thermal softening matter: proxies that reduce a friction parameter can reproduce trends but tend to overcorrect (HTS predicted an 18.9% reduction vs 12.7% measured), while coupled FS-OB and CID models closely matched observations by resolving temperature rise and feeding it into interface strength or capacity.

Our assessment emphasizes that inputs were taken from specimen-matched measurements and carried directly into



simulations, rather than retrospectively adjusted to align torque curves. Parameter-transfer residuals of 2.2–4.1% in frictional dissipation per turn across mechanisms indicate that pitch-wise energy matching between models is viable, and this underpinned the consistency observed across the three formulations. Against 76 insertions divided by mechanism, alignment metrics were within single digits for HTS in friction and pilot-hole mechanisms, mid-single digits for FS-OB in thermal and microcrushing, and high single digits for CID post-peak energy and in thermal coupling, with double-digit errors where CID was not tuned to per-regime friction.

Taken in aggregate, checks on the correctness of the numerical implementation supported reliance on the solvers for the questions asked: manufactured solutions and patch tests maintained errors below approximately 0.6%, symmetry checks were satisfied, and energy balances closed to within 0.5%. Refinement studies across solvers constrained numerical perturbations to approximately 1–3% for torque, which set the noise floor for interpreting model-to-experiment differences.

One useful by-product of a multi-mechanism assessment is the ability to see coupling that is not necessarily placed at center stage. Pilot-hole undersizing increases early forming work but also modestly shifts the seating ramp, and the models were consistent with this small lift. Thermal softening that lowers tangential traction in mid-advance can also change the onset of microcrushing by reducing bearing loads, and this interplay was apparent in FS-OB runs at 50 rpm, where the peak torque lagged by roughly 0.2 turns relative to 30 rpm conditions. Such couplings reinforce the value of the FS-OB model when multi-factor decision-making is required.

The workflow applied here resisted the easy temptation to manipulate parameters in a way that smears physics and confounds conclusions. Using measured  $\tau_0$ ,  $\delta f$ , and  $\beta$  values in CID without per-regime retuning forced the model to earn its agreement; in friction-dominated conditions, the penalty was a 12.7% MAPE. Conversely, in regimes for which the model was designed (post-peak), the benefit was a faithful reproduction of torque decay and damage work within 8.2%. This type of mechanism-aware calibration and comparison supports discriminating deployment of the models: sector models for rapid what-if exploration and hardware iteration; full-screw models when head seating and thermal feedback cannot be ignored; cohesive models when the task is to anticipate loss of fixation or study removal torques.

Beyond the per-row evidence tabulated, we draw eight cross-cutting conclusions with relevance to users of these models. First, the laboratory evidence employed in this assessment was fully traced to collection sites and lots, with independent replication at a second site reproducing mean peak torque within 6% across all mechanism-focused subsets. Second, transferring the experimental settings into analysis parameters via a single mapping protocol conserved mechanical work per pitch to within 4% between models depending on mechanism, demonstrating that parameters carried between HTS and FS-OB were effectively equivalent. Third, the retained geometry and contact relationships captured the energy-bearing paths observed in micro-CT and DIC—no load paths critical to torque

generation were blocked or omitted, which the 6% match in forming volume and the near-identical flank pressure gradients support. Fourth, the numbers assigned to friction, elastic moduli, and cohesive strengths came directly from measurements on the same or sister specimens and were not adjusted to fit torque curves; Monte Carlo draws propagated those input scatters to prediction intervals.

Fifth, manufactured-solution benchmarks and symmetry checks showed errors between 0.2% and 1.2% depending on module, indicating that code-level correctness was quantitatively established before any solver tuning. Sixth, changes in axial increment and contact stiffness produced torque variation generally below 2%, with the largest values (approximately 3%) appearing under seating nonlinearities, bounding numerical perturbations well below model-to-experiment differences. Seventh, across 76 insertions from two laboratories, quantitative alignment of torque–advance curves produced NRMSE values of 6.1% (HTS), 7.8% (FS-OB), and 9.3% (CID pre-peak; 10.5% including post-peak), and mechanism-specific MAPEs that were likewise within single digits for the models targeted to those mechanisms. Eighth, the analysis stack was operated under a documented engineering process with automated testing between 79% and 86% line coverage depending on module set, zero severity-1 open defects at release, and archived artifacts enabling exact reproduction of the runs reported here.

*Decision-readiness ladder — gradation.*

- a.  
Exploratory only: not for decisions.
- b.  
Inform design trends: qualitative decisions with margin.
- c.  
Support quantitative trade-offs: single-digit uncertainty on targeted mechanisms.
- d.  
Support verification and validation activities: multi-mechanism with audit trail and uncertainty budgets.

## 18. Limitations

This study did not systematically vary donor age, bone mineral density, or anatomical site beyond ovine tibial cortex, and while specimens were randomized across labs, a broader sampling would better represent clinical heterogeneity. The torque transducers used have high resolution, but thread cutting noise and occasional stick–slip spurts produced high-frequency content that was filtered; these details may matter in micro-mechanical interpretations. The HTS model’s thermal proxy is a crude representation for temperature-dependent behavior; adding a coupled heat equation would likely improve softening predictions without a prohibitive cost increase.

We did not explore screw reuse or progressive blunting in a way that would allow definitive statements about use history; likewise, our test samples focused on well-prepared blocks and did not include pathological cortex or prior damage states beyond the controlled microcrushing at seating. The CID model favored a mode-II-centric depiction of interfacial degradation;

combined normal–tangential coupling at the flank and mode mixity calibration could further refine post-peak response. Finally, while the software process achieved substantial test coverage, tests are inevitably partial, and new features can alter behavior subtly; continuous integration mitigates but does not eliminate such risks.

## 19. Conclusions

A mechanism-focused credibility assessment for three implicit FEA models of screw insertion torque was completed under a 0–7 CRI scale, producing 96 graded entries with quantitative justifications. The HTS model is decision-ready for friction control and pilot-hole strategy questions, the FS-OB model for seating and thermo-mechanical questions, and the CID model for anticipating degradation and post-peak torque decay. Parameters were mapped from experiment to analysis with minimal residuals; numerical correctness and solver robustness were demonstrated; and outputs aligned to laboratory measurements within single digits in targeted regimes. The software process supporting these conclusions was engineered, versioned, and tested with high coverage, sustaining reproducibility.

Future work will tighten the thermal representation in the HTS, extend the CID to coupled mode mixity, and expand datasets to include human cortical bone across a wider age and density range. With those extensions, the credibility demonstrated here should generalize to broader surgical and design contexts.

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## 21. Supplemental Material

The supplemental package contains: (i) anonymized raw torque, rotation, advance, and temperature time series for all 76 insertions; (ii) screw lot numbers, bone donor IDs, and full metadata; (iii) mesh convergence studies for each model with element sizes and torque deltas; (iv) details of manufactured-solution setups and code-check scripts; (v) additional figures comparing torque–advance curves by mechanism and model; and (vi) a step-by-step recipe to reproduce the FS-OB thermo-mechanical runs, including solver settings and boundary conditions.

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Table 1: Credibility Readiness Index (CRI, 0–7) for the Helical Thread Sector (HTS) model by mechanism and factor; one row per (factor × mechanism).

| Credibility factor                                                                                          | Level | Basis                                                                                                                         |
|-------------------------------------------------------------------------------------------------------------|-------|-------------------------------------------------------------------------------------------------------------------------------|
| Data pedigree - Helical Thread Sector (HTS) Model - Interfacial friction control                            | 6     | Friction subset n=24 across two labs; cross-lab peak torque means within 5.2%; calibrated transducers within 0.5% FS.         |
| Equivalency of input parameters - Helical Thread Sector (HTS) Model - Interfacial friction control          | 6     | Unified mapping gave 2.4% residual in per-turn dissipation versus FS-OB under matched $\mu$ .                                 |
| Model form - Helical Thread Sector (HTS) Model - Interfacial friction control                               | 6     | Periodic sector preserved flank shear load path; flank pressure gradients matched micro-CT-inferred forming volume within 6%. |
| Model inputs - Helical Thread Sector (HTS) Model - Interfacial friction control                             | 6     | Used specimen-matched values ( $\mu = 0.35 \pm 0.05$ dry; $0.20 \pm 0.04$ saline; $0.45 \pm 0.06$ roughened) without tuning.  |
| Numerical code verification - Helical Thread Sector (HTS) Model - Interfacial friction control              | 6     | Manufactured-solution and patch tests within 0.2–0.4%; helical invariance within 0.3%.                                        |
| Numerical solver error - Helical Thread Sector (HTS) Model - Interfacial friction control                   | 6     | Halving increment changed torque by +1.4%; halving penalty by +1.6%; tighter tolerances <0.4%.                                |
| Output comparison - Helical Thread Sector (HTS) Model - Interfacial friction control                        | 6     | MAPE 6.0% across dry/saline/roughened; peak torque within 0.07 N·m of means.                                                  |
| Software quality assurance - Helical Thread Sector (HTS) Model - Interfacial friction control               | 6     | 79% line coverage; 0 open severity-1 defects; archived binaries and inputs with hashes.                                       |
| Data pedigree - Helical Thread Sector (HTS) Model - Cortical microcrushing at thread root                   | 6     | Microcrushing subset n=18; cross-lab mean peak differences 5.9%; DIC strain data archived.                                    |
| Equivalency of input parameters - Helical Thread Sector (HTS) Model - Cortical microcrushing at thread root | 6     | Mapping residual 3.1% in frictional dissipation per turn relative to FS-OB near seating.                                      |
| Model form - Helical Thread Sector (HTS) Model - Cortical microcrushing at thread root                      | 5     | Capped compressive yield proxy under-represented seating ramp; DIC root strain matched within 10%.                            |
| Model inputs - Helical Thread Sector (HTS) Model - Cortical microcrushing at thread root                    | 6     | Cortex yield $105 \pm 15$ MPa, hardening $0.85 \pm 0.25$ GPa used as measured; no posterior tuning.                           |
| Numerical code verification - Helical Thread Sector (HTS) Model - Cortical microcrushing at thread          | 6     | Patch and symmetry tests as above; energy balance within 0.5% per cycle.                                                      |

Table 2: Credibility Readiness Index (CRI, 0–7) for the Full-Screw Orthotropic Bone Block (FS-OB) model by mechanism and factor.

| Credibility factor                                                                                               | Level | Basis                                                                                        |
|------------------------------------------------------------------------------------------------------------------|-------|----------------------------------------------------------------------------------------------|
| Data pedigree - Full-Screw Orthotropic Bone Block (FS-OB) Model - Interfacial friction control                   | 6     | Friction subset n=24; two labs; peak torque means within 5.2%; calibration within 0.5% FS.   |
| Equivalency of input parameters - Full-Screw Orthotropic Bone Block (FS-OB) Model - Interfacial friction control | 6     | Mapping residual 2.8% in per-turn dissipation vs HTS under matched $\mu$ .                   |
| Model form - Full-Screw Orthotropic Bone Block (FS-OB) Model - Interfacial friction control                      | 6     | Full helix and head seating preserved load paths; orthotropy modeled; slip bands reproduced. |
| Model inputs - Full-Screw Orthotropic Bone Block (FS-OB) Model - Interfacial friction control                    | 6     | $\mu$ from tribometry applied (0.20/0.35/0.45); orthotropic E from specimen tests.           |
| Numerical code verification - Full-Screw Orthotropic Bone Block (FS-OB) Model - Interfacial friction control     | 5     | Manufactured-solution and patch errors 0.4–0.6%; energy check within 0.5%.                   |
| Numerical solver error - Full-Screw Orthotropic Bone Block (FS-OB) Model - Interfacial friction control          | 6     | Increment halving +1.1%; penalty halving +1.7%; tighter tolerances <0.5%.                    |
| Output comparison - Full-Screw Orthotropic Bone Block (FS-OB) Model - Interfacial friction control               | 5     | MAPE 8.9% across friction regimes; peaks within 0.09 N·m.                                    |
| Software quality assurance - Full-Screw Orthotropic Bone Block (FS-OB) Model - Interfacial friction control      | 7     | 86% coverage; 0 severity-1, 1 severity-2 open defect; archived artifacts and reviews.        |
| Data pedigree - Full-Screw Orthotropic Bone Block (FS-OB) Model - Cortical microcrushing at thread root          | 6     | Microcrushing subset n=18; DIC strains archived; cross-lab differences 5.9%.                 |
| Equivalency of input parameters - Full-Screw Orthotropic Bone Block (FS-OB) Model - Cortical microcrushing at    | 6     | Residual 3.5% in dissipation mapping vs HTS near seating.                                    |

Table 3: Credibility Readiness Index (CRI, 0–7) for the Cohesive Interface Damage (CID) model by mechanism and factor.

| Credibility factor                                                                                              | Level | Basis                                                                                                                              |
|-----------------------------------------------------------------------------------------------------------------|-------|------------------------------------------------------------------------------------------------------------------------------------|
| Data pedigree - Cohesive Interface Damage (CID) Model - Interfacial friction control                            | 6     | Friction subset n=24; two-lab replication; calibrated transducers; raw files archived.                                             |
| Equivalency of input parameters - Cohesive Interface Damage (CID) Model - Interfacial friction control          | 5     | Cohesive $\tau_0$ chosen to span $\mu$ regimes; dissipation residual vs HTS 3.0%.                                                  |
| Model form - Cohesive Interface Damage (CID) Model - Interfacial friction control                               | 5     | Cohesive flanks plus normal friction capture shearing; no head seating shortfall.                                                  |
| Model inputs - Cohesive Interface Damage (CID) Model - Interfacial friction control                             | 5     | $\tau_0 = 35 \pm 6$ MPa; $\delta f = 15 \pm 3$ $\mu$ m; $\tau_r = 0.15 \tau_0$ ; applied per tribometry without per-regime tuning. |
| Numerical code verification - Cohesive Interface Damage (CID) Model - Interfacial friction control              | 5     | Mode II cohesive benchmark recovered fracture energy within 1.2%.                                                                  |
| Numerical solver error - Cohesive Interface Damage (CID) Model - Interfacial friction control                   | 5     | Increment halving +1.6%; penalty halving +2.0%; tolerances <0.6%.                                                                  |
| Output comparison - Cohesive Interface Damage (CID) Model - Interfacial friction control                        | 4     | MAPE 12.7% across friction regimes; peaks within 0.12 N·m.                                                                         |
| Software quality assurance - Cohesive Interface Damage (CID) Model - Interfacial friction control               | 6     | 80% coverage; 0 severity-1 open defects; archived artifacts.                                                                       |
| Data pedigree - Cohesive Interface Damage (CID) Model - Cortical microcrushing at thread root                   | 6     | Microcrushing subset n=18; DIC and torque synchronized; cross-lab delta 5.9%.                                                      |
| Equivalency of input parameters - Cohesive Interface Damage (CID) Model - Cortical microcrushing at thread root | 5     | Mapping residual 3.7% vs FS-OB in dissipation near seating.                                                                        |
| Model form - Cohesive Interface Damage (CID) Model - Cortical microcrushing at thread root                      | 6     | Cohesive degradation reorganized load sharing post-peak; captured torque drop.                                                     |
| Model inputs - Cohesive Interface Damage (CID) Model - Cortical microcrushing at thread root                    | 5     | Cohesive parameters from tribometry; cortex E from specimens; no torque tuning.                                                    |
| Numerical code                                                                                                  | 5     | Cohesive benchmark 1.2% er-                                                                                                        |